



ADICHUNCHANGIRI UNIVERSITY
B G S Institute of Technology
 B G Nagar, Nagamangala Tq, Mandya Dist.
Department of Electronics and Communication
Engineering



Question Bank of Python Programming (22EC641)

Module 1

1. What is Python programming? List & explain the important features and applications of the Python programming. 10M
2. Differentiate between compiler and interpreter. 06M
3. List and explain values & types of the python with examples. 06M
4. Discuss types of errors in the Python Programming? 06M
5. Predict the output of the following code and justify your answer. 10M

i.	<pre>>>>x='23' >>>y=13 >>>print(x+y)</pre>	ii.	<pre>>>>x=7 >>>y=sqrt(x) >>>print(y)</pre>
iii.	<pre>>>>state="Karnataka" >>>print(state[2]) >>>state[1]=state[4]="a" >>>state[7]="t" >>>print(state)</pre>	iv.	<pre>>>>def f(x): y=2*x return x >>> x=1 >>> x=f(x+1) >>> print(x) >>>print(y)</pre>
v.	<pre>>>>print(bool("False")) >>>print(bool())</pre>		

6. List and explain the most commonly used built-in types of python with examples. 10M
7. Write a note on: 10M
 - (i) Python Keywords
 - (ii) Comments in python
 - (iii) Order of Operation
8. Write a python program to calculate the area of a circle and the circumference of a circle. Take the inputs from the keyboard and print the results. 6M
9. What are functions? Explain any two built-in functions, math functions and random functions. 10M
10. Discuss types of errors in the Python Programming? Explain the rules to naming of Identifiers and identify the valid and invalid identifier in the given list: 10M

(i) python_641	(ii) import python641
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(iii) 641_Python	(iv) _Python641
(v) python641#	(vi) python_ _641

11. Discuss how to define and call the functions. Illustrate with a user defined function example. 10M
12. Explain the flow of execution in python and differentiate between arguments and parameter. Draw and explain stack diagram by taking an example function. 10M

Module-2

1. Discuss various operators used in python and evaluate the output for the following expressions: 10M
- (i) 54//7 (ii) -31%8 (iii) 0 **and** 7 (iv) 0 **or** 7
(v) 9 **and** 7 (vi) 9 **or** 7 (vii) 7>=3 (viii) 7 **is not** 7.
2. How do you compare two strings in Python? Illustrate with example. Write Python program to count the total number of vowels, consonants and blanks in a String. 10M
3. What is Iteration? Explain while and for loop. Write a python program to check whether the given number is prime or not using while loop. 08M
4. Discuss the working of range() function. Write a python program to demonstrate counting, summing, and an average of elements using loops. 08M
5. Write a python program to prompt for a score between 0 to 100. If the score is out of range, print an error message. If the score is between 0 to 100, print a grade using the following table: 8M

Score	Grade
>=90	A
>=70	B
>=50	C
>=35	D
<35	F

6. Explain break, continue and pass statements in loop structure with examples. Find the sum of any 5 odd numbers taken as input from the keyboard. 10M
7. Discuss string type in python. Write a python program to demonstrate traversal through a string with a for loop and while loop. 10M
8. List and explain any 5 string Methods with an example for each. 10M

9. If S= **“BGS Institute of Technology”** Write a Python program to count the number of characters and display the output of how many times the letter ‘t’ appears in that text. 8M
10. If S= **“Python Programming”** Write a Python program to counts the number of Characters and how many times the letter ‘m’ appears in that text. 8M
11. Develop a Python Program to Display the multiplication Tables and Find the sum of any 5 odd numbers taken as input from the keyboard. 5M
12. Explain the concept of conditional execution, alternative execution, chained conditionals, and nested conditionals with suitable examples. 8M
13. Draw and explain stack diagram by taking an example function. Differentiate between void and fruitful function. 7M

Module-3

1. Discuss the list in python? Explain the concept of list slicing and list traversing for the example below: 12M

Let S= *“Python Programming”*

L =list(S) and Find:

i) <i>L[0:6]</i>	ii) <i>L[3:-1]</i>
iii) <i>L[6:]</i>	iv) <i>L[: :-1]</i>
v) <i>L[: :-2]</i>	vi) <i>L[: : 3]</i>
vii) <i>L[-1 1:-4]</i>	viii) <i>L[1:3]='i','j' print(L)</i>

2. Make a list of all the letters of the alphabet then using the slice operation do the following: 8M
 - (i) Print the first three letters from the list
 - (ii) Print any three letters from the middle
 - (iii) Print the letters from 5th letter to the end of the list
 - (iv) Print the letters in reverse order
 - (v) Print the even indexed letters
3. Describe any 8 list methods with examples. 8M
4. Write a python program to sort numbers in a list in ascending order by passing the list as an argument to the function call. 6M
5. Discuss the dictionary type in python. Explain the dictionary as a collection of counters and its implementation for the string “python programming” 10M

6. Explain Dictionaries and Lists with a function that invert a dictionary along with the state diagram. Write a Python program to generate a dictionary that contains **(i: i*i)** such that **i** is a number ranging from 1 to n. 8M

7. Explain the concepts
 - i. Reverse lookup 8M
 - ii. Memos

8. Consider the list scores = [3, 7, 4, 5, 8, 1, 9] and write the python code to perform the following operations: 6M
 - i) Insert an element 2 at the beginning of the list.
 - ii) Insert an element 5 at the index position 3 of the list.
 - iii) Insert an element 6 at the end of the list.
 - iv) Delete an element at the beginning of the list
 - v) Delete an element at the index position 3.
 - vi) Delete all the elements of the list.

9. Discuss the difference between identical objects and equivalent object. How are they identified? give examples. 6M

10. Discuss the dictionary type in python with the key-value relationship and Explain keys (), values (), get () & items () Methods with examples. 6M

11. If S= "**Adichunchanagiri University**", Find:
 - i) S[0: :3] ii) S [:12] iii) S [17:] iv) S [::-1]
 and Write a Python program to count the number of characters and display the output of how many times the letter 'i' appears in that string S. 8M

Module 4

1. Define tuple. Explain the DSU pattern. Write Pythonic code to demonstrate tuples by sorting a list of words from longest to shortest using loops. 10M

2. How tuples can be used as a key for dictionaries? Discuss with example. 10M

3. Compare and contrast tuples with lists. Explain the following operations in tuples 12M
 - (i) Sum of two tuples
 - (ii) Slicing operators
 - (iii) Compression of two tuples

4. What is file handler? What is the common run-time error faced while opening a file? How do you handle it? Explain with example 10M

5. Discuss open (), close (), read (), write () function on files. 10M

Module 5

1. Discuss GUI & List down the various options of button widget. 8M

2. Explain the concept of a Graphical User Interface (GUI) and describe how Tk() and mainloop() function in a tkinter-based application. Illustrate with a basic tkinter example that creates a blank window. 10M

3. Write a Python program using tkinter to create a window titled "Welcome App"

- with fixed size 600×400 pixels. Explain how the event loop maintains responsiveness. 10M
4. Explain the purpose of widgets in tkinter. Create a login form with Label, Entry, and Button widgets, arranging them using the .grid() method. 10M
 5. Compare .pack(), .grid(), and .place() geometry managers in tkinter. Write code to demonstrate how two buttons can be placed using .pack() and .place() respectively. 10M
 6. What is the Canvas widget in tkinter? Write a program that creates a canvas of size 400×300 and draws a rectangle, a circle (oval), and a line with different colors. 10M
 7. Explain the process of creating a menu bar in tkinter. Write a program that adds a "File" menu with options "Open", "Save", and "Exit", and attach it to the main window. 10M
 8. Define what a callable is in Python. Write a class with a __call__() method that prints a message when used as a tkinter button callback. 10M
 9. Explain the concept of Numpy in detail. 10M
 10. Explain the concept of Pandas and Data Visualization. 10M
 11. You have a dataset that records monthly revenue of a company over the past year. The dataset includes the columns `Month` and `Revenue`. Using **Pandas** and **Matplotlib**, write a program to: 10M

Month	Revenue
Jan	120
Feb	130
Mar	140
Apr	160
May	180
Jun	190
Jul	200
Aug	210
Sep	220
Oct	230
Nov	250
Dec	260

- a) Plot the **monthly revenue** as a line chart.
- b) Plot a **bar chart** to show the total revenue for the first half and the second half of the year.
- c) Annotate the line chart with the highest revenue month and value.

ALL THE BEST